CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Case 10-2026: A 70-Year-Old Woman with a Racing Heart, Fatigue, and Dyspnea

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PRESENTATION OF CASE

Dr. Eliza D. Hompe: A 70-year-old woman was evaluated at this hospital because of a racing heart, fatigue, dyspnea, and leg edema.

Five months before the current admission, the patient was evaluated by a cardiologist at this hospital for frequent episodes of palpitations and lightheadedness that did not resolve with cessation of alcohol and caffeine use. Outpatient rhythm monitoring showed paroxysmal atrial tachycardia and atrial fibrillation events that constituted a total tachycardia burden of 6%, with heart rates of up to 180 beats per minute. Treatment with apixaban and flecainide was started; however, the symptoms did not abate, and flecainide was discontinued.

One month before the current admission, percutaneous catheter ablation was performed with the patient under general anesthesia. The procedure included pulsed-field ablation for pulmonary-vein isolation and posterior-wall isolation, as well as radiofrequency ablation in the coronary sinus, the slow pathway of the atrioventricular node, and the basal septum because ectopic beats from these sites had been observed to provoke paroxysmal atrial fibrillation. The patient was discharged home on the same day.

The next day, the patient called her cardiologist to report 2.0 kg of weight gain since the procedure and new edema in both legs. Oral furosemide was prescribed. Eleven days later, the patient called her cardiologist to report intermittent episodes of palpitations and sharp, “cramping” pain on the left side of the chest that were accompanied by a sensation of heartburn; ibuprofen was prescribed. The next day, myalgias in the neck, arms, and torso developed, along with a mild nonproductive cough. Point-of-care testing for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection was positive, and the patient took acetaminophen for symptom relief.

Sixteen days after the ablation procedure and 2 weeks before the current admission, the patient had extreme fatigue, persistent pedal edema, increasing dyspnea occurring with minor exertion, and an additional 2.8 kg of weight gain. She contacted

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Table 1. Laboratory Data.*

Variable	Reference Range, Adults, Other Hospital	10 Days before Current Admission, Other Hospital	Day of Current Admission, Other Hospital	Reference Range, Adults, This Hospital†	On Current Admission, This Hospital
Hemoglobin (g/dl)	11.2–15.7	10.1	12.3	12.0–16.0	12.0
Hematocrit (%)	34.1–44.9	31.4	37.9	36.0–46.0	37.9
White-cell count (per μl)	4000–10,000	13,100	16,200	4500–11,000	12,390
Differential count (per μl)					
Neutrophils	—	—	—	1800–7700	10,900
Lymphocytes	—	—	—	1000–4800	740
Platelet count (per μl)	182,000–369,000	413,000	507,000	150,000–400,000	325,000
Sodium (mmol/liter)	136–145	129	132	135–145	135
Potassium (mmol/liter)	3.5–5.1	4.4	4.6	3.4–5.0	4.8
Chloride (mmol/liter)	98–107	98	102	98–108	98
Carbon dioxide (mmol/liter)	20–31	21	20	23–32	17
Urea nitrogen (mg/dl)	9–23	34	34	8–25	29
Creatinine (mg/dl)	0.55–1.52	0.71	1.11	0.60–1.50	1.10
Glucose (mg/dl)	74–106	106	177	70–110	121
Albumin (g/dl)	3.4–5.0	3.0	2.8	3.3–5.0	3.2
Alanine aminotransferase (U/liter)	7–40	170	71	7–33	53
Aspartate aminotransferase (U/liter)	13–40	71	44	9–32	45
Alkaline phosphatase (U/liter)	46–116	451	301	45–115	280
Total bilirubin (mg/dl)	0.2–1.1	1.7	1.6	0.0–1.0	1.2
Troponin I (ng/liter)	<45	7	7	—	—
B-type natriuretic peptide (pg/ml)	<100	196	518	—	—
Lactate (mmol/liter)	0.5–1.6	—	4.3	0.5–2.2	2.7
Prothrombin time (sec)	—	—	—	10.0–13.0	27.8
International normalized ratio	—	—	—	0.9–1.1	2.5
Partial-thromboplastin time (sec)	—	—	—	24.0–37.5	26.4

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for lactate to milligrams per deciliter, divide by 0.1110.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

her primary care physician and was subsequently admitted to another hospital with a diagnosis of heart failure.

On evaluation at the other hospital, the patient was afebrile, with a temporal temperature of 36.6°C. The heart rate was 90 beats per minute and regular, but paroxysmal atrial fibrillation occurred with a heart rate of up to 140 beats per minute. The blood pressure was 179/81 mm Hg, the respiratory rate 27 breaths per minute, and the oxygen saturation 95% while the patient was breathing ambient air. Laboratory test results are shown in Table 1. An electrocardiogram (ECG) reportedly showed sinus rhythm, PR-interval depression, and inferolateral ST-segment elevation.

Dr. Jo-Anne O. Shepard: A chest radiograph (Fig. 1A) showed moderate bilateral pleural effusions and enlargement of the cardiac silhouette.

Dr. Hompe: Echocardiography showed normal ventricular function, a small left ventricular cavity, pleural effusions, moderate pericardial effusion, and a dilated inferior vena cava. Oral amiodarone, oral colchicine, intravenous furosemide, intravenous and oral metoprolol, and intravenous diltiazem were administered. After 2 days, the patient was maintaining sinus rhythm and had lost 3.0 kg of weight; she was discharged from the hospital.

Five days later, the patient stopped taking colchicine because of diarrhea. She had a telehealth visit with her cardiologist and reported that the dyspnea on exertion had abated somewhat, but she also reported difficulty and pain with swallowing.

Three days later, on the day of the current admission, the patient was evaluated in the emergency department of the other hospital for marked dyspnea with the inability to catch her breath, a racing heart, and fatigue. She was afebrile, with a temporal temperature of 36.4°C. The heart rate was 131 beats per minute and irregular, the blood pressure 131/92 mm Hg, the respiratory rate 26 breaths per minute, and the oxygen saturation 95% while she was breathing ambient air. The examination revealed decreased breath sounds at the lung bases and edema in the legs. Laboratory test results are shown in Table 1.

Dr. Shepard: A chest radiograph obtained on admission to the other hospital (Fig. 1B) showed new pneumopericardium with pericardial thickening and persistent pleural effusions.

Dr. Hompe: Blood was obtained for culture, and empirical treatment with intravenous vancomycin and piperacillin–tazobactam was administered. The patient was transferred to the cardiac intensive care unit (ICU) of this hospital.

On admission to the cardiac ICU, review of systems was notable for leg edema, dyspnea, and anorexia. The patient reported that difficulty swallowing had started after the ablation procedure. She reported no fever, chills, emesis, nausea, abdominal pain, chest pain, bleeding, or dizziness.

The patient's medical history included carotid artery disease (for which she had undergone right carotid endarterectomy), hyperlipidemia, torticollis (for which she had received botulinum toxin injections), osteoarthritis (for which she had undergone right hip arthroplasty), and anxiety. Medications included apixaban, amiodarone, colchicine, metoprolol, pravastatin, and famotidine, as well as acetaminophen and diazepam as needed. Sulfonamides had caused nausea and vomiting. She had received vaccinations against SARS-CoV-2.

The patient lived in a coastal region of Massachusetts with her husband and was retired. Her family history was notable for prostate cancer and coronary artery disease in her father and a fatal myocardial infarction in her paternal grandfather. She had quit smoking cigarettes more than 50 years earlier and did not use alcohol or other substances.

On examination, the temporal temperature was 36.6°C, the heart rate 113 beats per minute and irregular, the blood pressure 163/77 mm Hg, and the oxygen saturation 94% while the patient was breathing ambient air. She appeared anxious. Auscultation of the chest revealed irregular tachycardia, a friction rub, tachypnea, and diminished basilar breath sounds. Edema in the legs was noted.

Laboratory test results are shown in Table 1. Testing for SARS-CoV-2 RNA was negative. Chest radiography and ECG were performed.

Dr. Shepard: A chest radiograph obtained on admission to this hospital (Fig. 1C) showed findings consistent with those seen at the other hospital.

Dr. Shaan S. Khurshid: In contrast to an ECG obtained immediately after percutaneous catheter ablation (Fig. 1D), an ECG obtained on the current admission (Fig. 1E) showed atrial fibrillation with a rapid ventricular response, nonspecific ST-segment changes, and changes consistent with low QRS voltage in the precordial leads.

Dr. Hompe: A diagnosis and management decisions were made.

DIFFERENTIAL DIAGNOSIS

Dr. Thomas K. Varghese, Jr.: This 70-year-old woman with a long-standing coexisting condition (atrial fibrillation) had recently undergone an intervention (percutaneous catheter ablation) and did not receive the intended benefit. Instead, the day after her procedure, she had evidence of fluid retention, which began a cascade of symptoms in the subsequent weeks that culminated in admission to the cardiac ICU approximately 4 weeks after the procedure.

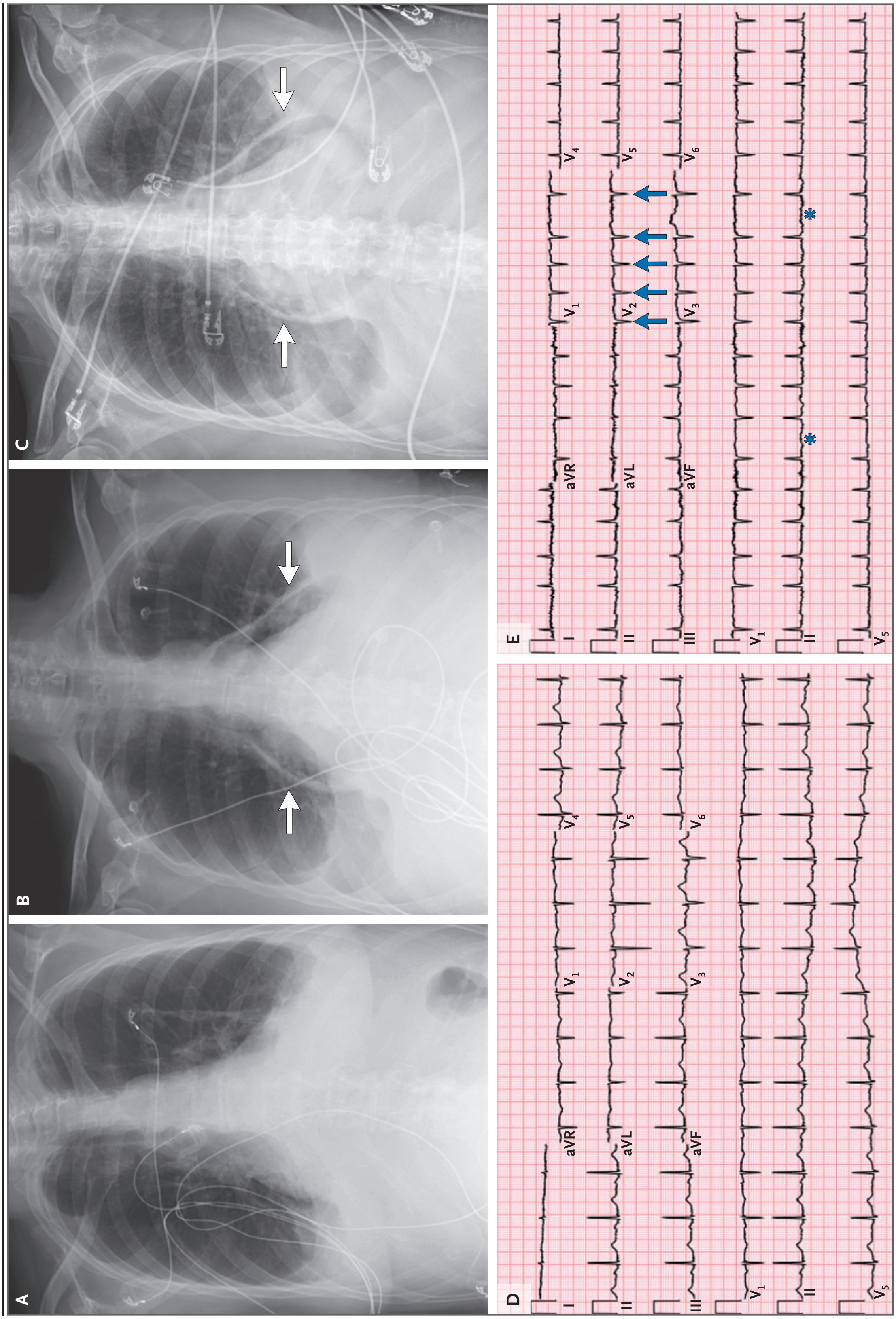
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Figure 1 (facing page). Initial Chest Radiographs and Electrocardiograms.

Percutaneous catheter ablation was performed 1 month before the current admission. A chest radiograph obtained 20 days after the procedure (Panel A) shows moderate pleural effusions. Chest radiographs obtained 28 days after the procedure (Panels B and C), on the day of the current admission, show new pneumopericardium with diffuse pericardial thickening (arrows). A 12-lead electrocardiogram obtained immediately after the ablation procedure (Panel D) shows sinus rhythm and normal findings overall. A 12-lead electrocardiogram obtained 28 days after the procedure (Panel E), on the day of the current admission, shows atrial fibrillation with a rapid ventricular response, as evidenced by an undulating baseline (blue asterisks) with irregularly irregular RR intervals. As compared with the tracing obtained 1 month earlier, this tracing shows that the QRS amplitude no longer exceeds 1 mV across leads V_1 through V_6, including lead V_2 (blue arrows). Low QRS voltage, particularly when it develops over a relatively short time frame (days or weeks), may indicate fluid accumulation in the pericardial or pleural space.

My immediate concern is that the patient had a complication after percutaneous catheter ablation, which seems to be the inciting event. Postablation complications can include acute vascular injury (bleeding or stroke), cardiac perforation (cardiac tamponade), pulmonary-vein stenosis, heart failure, new arrhythmias, and damage to nearby structures such as the esophagus, vagus nerve, or phrenic nerve. Because the patient is presenting a month after her procedure, our list narrows to complications that are not immediately evident during the procedure.

The patient is afebrile, has an elevated heart rate and increased blood pressure, and reports dysphagia. An ECG shows atrial fibrillation and low QRS voltage in the precordial leads. Chest radiography shows air in the pericardium, an enlarged heart, and pleural effusions, with more fluid present on the left side than on the right. In developing a differential diagnosis to explain this patient's presenting symptoms, I will consider six possible diagnoses: heart failure syndrome, esophageal perforation, barotrauma, paraesophageal hernia, atrioesophageal fistula, and esophageal–pericardial fistula.

HEART FAILURE SYNDROME

When this patient was first admitted to the other hospital, her symptom complex included fatigue, pedal edema, worsening dyspnea on exertion, and weight gain. All these signs and symptoms are consistent with heart failure. However, the development of difficulty swallowing, which had occurred after her procedure, and the findings on chest radiography performed on transfer to this hospital are not consistent with a diagnosis of heart failure.

ESOPHAGEAL PERFORATION

Esophageal perforation is a rare but well-known complication of ablation procedures and should be considered as a possible diagnosis in this patient, especially given her difficulty swallowing. In patients with perforation of the esophagus, chest imaging is often notable for bilateral pleural effusions, which were present in this patient. However, patients also typically have sepsis, which did not appear to be an initial feature of this patient's case.

BAROTRAUMA

Barotrauma is another uncommon but well-known complication of ablation procedures. Barotrauma typically causes a clinically significant pneumothorax, which is usually easy to identify on chest imaging and was not noted in this patient. Furthermore, symptoms stemming from the pneumothorax, such as shortness of breath, would develop during or immediately after the procedure.

PARAESOPHAGEAL HERNIA

A paraesophageal hernia can appear on chest imaging as a retrocardiac mass or opacity, which may contain an air–fluid level and, sometimes, a gas-filled lucency. In this patient, a chest radiograph obtained 20 days after the ablation procedure showed moderate bilateral pleural effusions and an enlarged cardiac silhouette, and the radiograph obtained on the current admission, 8 days later, showed evidence of progression. In the absence of trauma, a paraesophageal hernia would not be associated with such rapid progression on imaging and is thus an unlikely diagnosis in this patient.

ATRIOESOPHAGEAL FISTULA

An atrioesophageal fistula often appears 2 to 6 weeks after an ablation procedure. However, the symptoms are rather severe and include chest pain, dysphagia, hematemesis, and melena. Although this patient had difficulty swallowing, she had no other symptoms at the time of the current evaluation that were suggestive of an atrioesophageal fistula. However, it is important to consider this diagnosis because it is a medical emergency associated with high mortality.

ESOPHAGEAL–PERICARDIAL FISTULA

Finally, an esophageal–pericardial fistula stemming from the procedure should be considered. The patient's clinical presentation was notable for a mix of esophageal and cardiac symptoms, particularly shortness of breath, difficulty swallowing, and air in the pericardial sac — the classic sign of an esophageal–pericardial fistula. On the basis of these findings, I think that the most likely diagnosis in this patient is an esophageal–pericardial fistula. I would recommend that contrast-enhanced computed tomography (CT) of the chest and abdomen be performed to establish the diagnosis.

CLINICAL IMPRESSION

Dr. Khurshid: Esophageal perforation with or without a fistula to the atrium (atrioesophageal fistula) or pericardium (esophageal–pericardial fistula) is a rare complication of ablation for atrial fibrillation that is performed with radiofrequency or cryothermal energy, with an incidence of 0.10 to 0.25%.[1] The risk of esophageal injury stems from the anatomical proximity of the esophagus to common ablation targets in the left atrium, right atrium, and coronary sinus (Fig. 2). Symptoms are often nonspecific and may include fatigue, malaise, fever, nausea, chest discomfort, dysphagia, odynophagia, hematemesis, and neurologic deficits. Because perforation and subsequent fistulization appear to develop in the context of evolving esophageal ulceration, the onset of symptoms is somewhat delayed, typically occurring 1 to 6 weeks after an ablation procedure.

Mortality associated with esophageal perforation after ablation is 80 to 90% without surgical correction, and therefore a high index of suspicion and prompt consultation with cardiothoracic surgery are key.[1] Initiation of broad-spectrum antibiotic therapy is recommended, given the potential for systemic infection. Until an atrioesophageal fistula has been ruled out, esophageal manipulation and instrumentation — including transesophageal echocardiography and upper endoscopy — are contraindicated because such procedures may lead to massive air embolism. It is notable that pulsed-field ablation is an emerging mode of energy delivery for cardiac ablation that appears to be selective for myocardial tissue; in recent reports that included more than 17,000 procedures, it was not associated with clinically apparent esophageal injury.[2]

In this patient's case, given the combination of symptoms that could be attributed to an esophageal condition (e.g., dysphagia) and signs on imaging that were compatible with pericardial involvement (e.g., pneumopericardium), along with the timing relative to the ablation procedure involving radiofrequency energy, we assumed a diagnosis of esophageal injury with a fistula until proven otherwise. Contrast-enhanced CT was performed to establish the diagnosis.

CLINICAL DIAGNOSIS

Esophageal–pericardial fistula.

DR. THOMAS K. VARGHESE, JR.'S DIAGNOSIS

Esophageal–pericardial fistula.

IMAGING STUDIES

Dr. Shepard: Contrast-enhanced CT performed at the other hospital on the day of admission to this hospital (Fig. 3) revealed moderate bilateral pleural effusions and moderate hydropneumopericardium with diffuse pericardial thickening. There was circumferential thickening of the esophagus adjacent to the left atrium, at the level of the superior and inferior pulmonary veins as they entered the left atrium.

DISCUSSION OF MANAGEMENT

Dr. Uma M. Sachdeva: The patient's imaging findings and clinical presentation were indicative of an esophageal–pericardial fistula with or without connection to the left atrium. No air was visualized within the left atrium on CT, and the patient had no neurologic symptoms suggestive of an active fistula to the left atrium. A fistula between the left atrium or pericardium and the esophagus can develop after thermal injury to the esophagus, when such injury leads to esophageal ulceration that evolves to perforation and subsequent fistulization. Erosion begins at the injured esophageal mucosa and progresses outward toward the left atrium, involving the pericardium, pulmonary veins, or left atrium. Esophageal injury is more common with ablation procedures that involve general anesthesia because of reduced esophageal motility, possible prolonged contact of the catheter with tissue, and

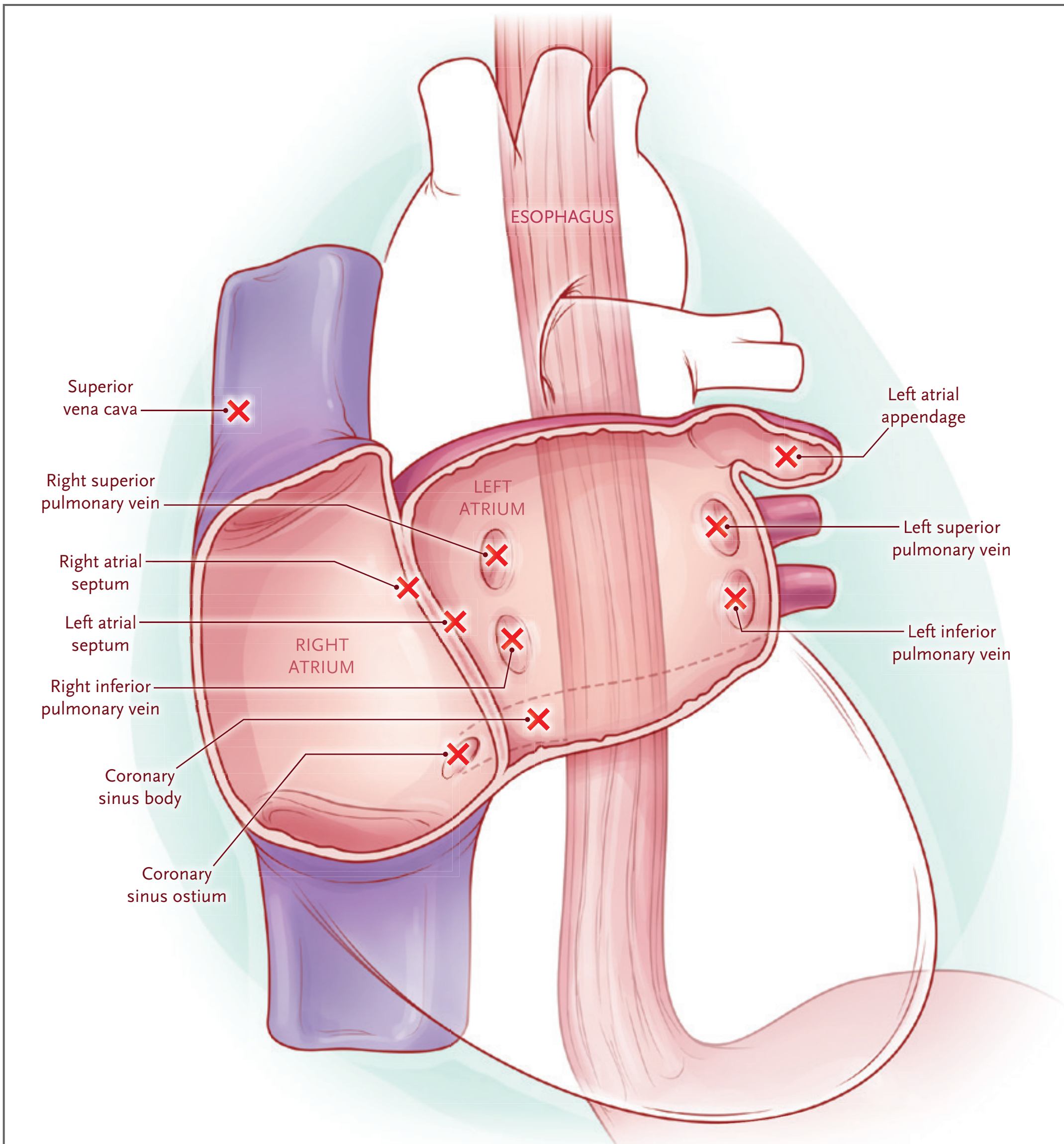

Figure 2. Relationship between the Esophagus and Targets Used in Ablation for Atrial Fibrillation.

Shown are common targets (red X symbols) used in ablation for atrial fibrillation and their typical anatomical relationship with the usual course of the esophagus in an approximate left anterior oblique projection. The relationship of the esophagus to the left atrium varies substantially, particularly within the coronal plane. Factors that affect heating of the esophagus (radiofrequency) or cooling of the esophagus (cryotherapy) are multiple and complex. In addition to anatomical proximity, such factors include pericardial fat volume; the number, duration, and sequence of energy applications; contact force; and catheter stability.

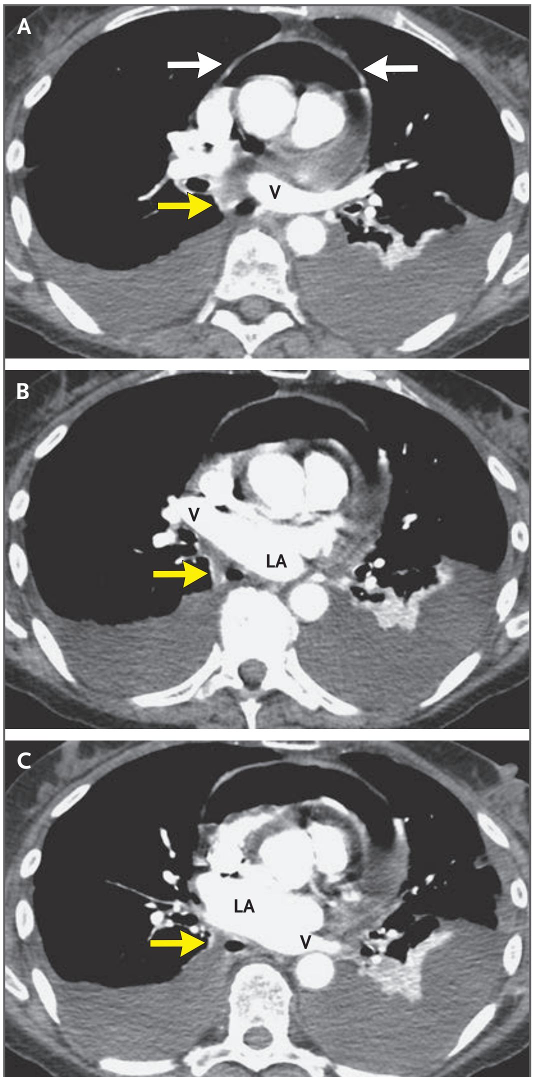

Figure 3. Chest CT Images.

Chest CT images obtained after the administration of intravenous contrast material show moderate layering effusions and basilar atelectasis. Moderate hydropneumopericardium (Panel A, white arrows) is present with diffuse pericardial thickening. Circumferential thickening of the esophagus (Panels A, B, and C; yellow arrows) is present adjacent to the left atrium (LA), at the level of the superior and inferior pulmonary veins (V) as they enter the left atrium. No air is visible within the cardiac chambers or in the mediastinum.

the inability of the patient to express discomfort during the procedure.

Among patients with an atrioesophageal fistula after transcatheter ablation, the prognosis remains guarded. A large meta-analysis involving 219 patients showed an overall survival of 47.0%.[3] Survival was 71.9% among patients who underwent surgical repair, as opposed to 11.0% among those who did not undergo surgery. Among patients who survived after surgery, the intervention included patch repair of the left atrium (in 61.1%), primary repair of the esophagus (in 68.3%), or repair with a tissue interposition flap placed between the esophagus and the left atrium (in 84.6%). Although multiple surgical approaches are available, right thoracotomy was performed in 45.1% of patients who survived after surgery. In addition, cardiopulmonary bypass was used in 84.8%.

Right thoracotomy allows for the best exposure of the intrathoracic esophagus and is the most common surgical approach used for an atrioesophageal fistula or esophageal–pericardial fistula. Left thoracotomy allows for the best access to the left inferior pulmonary vein, which is the most common site of fistulization.[4,5] Median sternotomy provides the best access for central cannulation in cardiopulmonary bypass; however, esophageal exposure is limited with this approach, and a two-stage procedure is indicated. For the prevention of embolic stroke from carbon dioxide or air insufflation, it is critical to avoid upper endoscopy until there is no connection between the esophagus and the left atrium or until an aortic cross-clamp is applied for cardiopulmonary bypass.

This patient underwent surgical repair performed by a team of surgeons from the thoracic surgery and cardiac surgery services. She underwent intubation with a double-lumen endotracheal tube, which allows for single-lung ventilation, and she was placed in the left lateral decubitus position. Transesophageal echocardiography was not performed, given the possibility of an esophageal fistula. Before the chest was entered, femoral arterial and venous sheaths were placed in the right groin so that cardiopulmonary bypass could be initiated if needed. Right thoracotomy was performed, and the fifth intercostal muscle was harvested on entry for later use in buttressing the esophageal repair. No purulent material within the chest was noted, and there was no obvious fistula, although the esophagus was adherent to the pericardium at the level of the inferior pulmonary vein. The esophagus was mobilized proximally and distally to this region; however, further dissection of the presumed fistula site was precarious because of dense adhesion. The pericardium was therefore opened for identification of the fistula tract.

Dr. Eriberto Michel: When the pericardium was entered anterior to the right phrenic nerve, abundant purulent fluid was immediately encountered. Samples of the pericardial fluid were sent for microbiologic culture, which would inform subsequent antimicrobial therapy. The pericardial sac was opened widely and irrigated with a copious amount of antibiotic and antifungal solution. Fibrinous rind was removed from both the visceral and the parietal portions of the pericardium. On inspection of the left atrium, there was no active fistula to the heart. At this point, esophagoscopy was performed to identify the location of the fistula to the pericardial sac. Endoscopy showed a 2-cm defect in the esophageal wall (Fig. 4A). The pericardial sac was filled with saline. With carbon dioxide insufflation of the esophageal lumen, bubbles were seen at the inferior aspect of the pericardium, indicating the fistula site. After the location of the esophageal–pericardial fistula was identified, the fistula was divided. The entire posterior left atrium was inspected, and an area of attenuated tissue was identified near the entry of the left inferior pulmonary vein. Primary repair at this site was performed with polypropylene sutures and then buttressed with pericardial tissue. Drains were placed in the inferior and anterior aspects of the pericardial sac.

Dr. Sachdeva: Once the fistula was divided, the esophageal defect was addressed (Fig. 4B). Primary repair was performed in two layers and then buttressed with an intercostal muscle flap placed between the esophagus and the pericardium. Before the esophagus was closed, a nasogastric tube was placed for intraluminal drainage. Additional drains were placed throughout the right side of the chest. Laparotomy was then performed for placement of a gastrojejunal drainage and feeding tube, with the gastric port to be used for drainage and the jejunal port for enteral feeding. The patient was in critical condition with continued intubation, and she was taken to the surgical ICU for postoperative treatment.

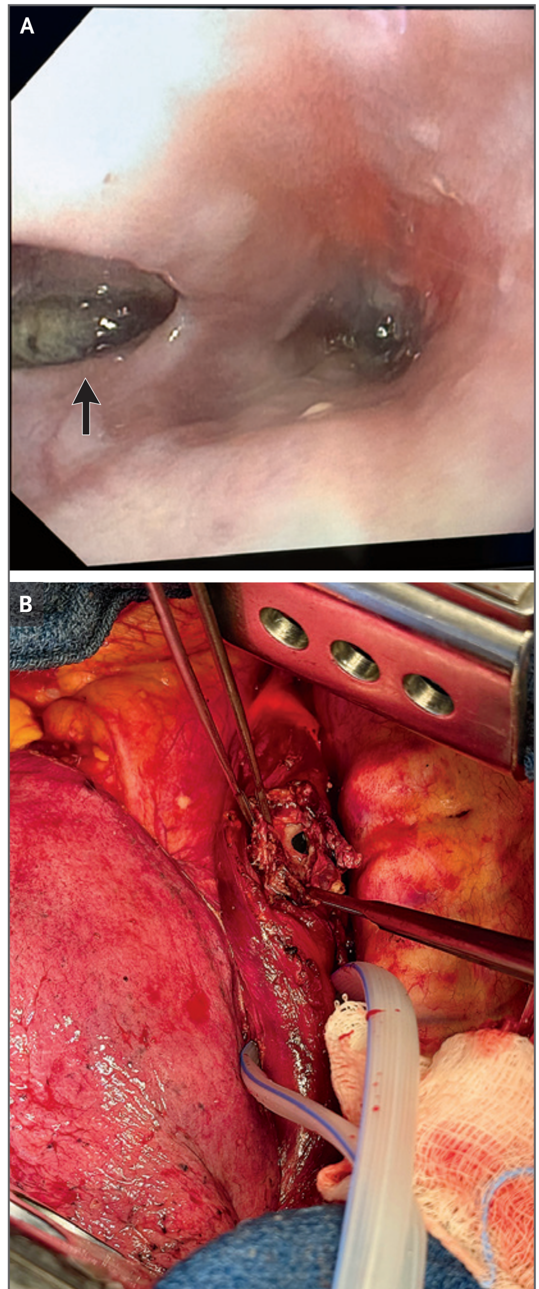

Figure 4. Intraoperative Images.

An intraoperative endoscopic image, obtained after a fistula to the left atrium or pulmonary veins was ruled out, shows a clear fistula from the esophageal lumen to the pericardial sac (Panel A, arrow). An intraoperative photograph, obtained once the esophagus was separated from the pericardium, shows a 2-cm defect involving the full thickness of the esophageal wall with a fibrotic rind (Panel B).

LABORATORY TESTING

Dr. Hompe: When the patient arrived at this hospital, empirical treatment including intravenous vancomycin, piperacillin–tazobactam, and fluconazole was administered; blood had been sent for culture. Broad-spectrum antibiotic therapy was initiated to provide coverage of oropharyngeal flora, given the clinical suspicion of esophageal perforation with a possible fistula to the pericardium or left atrium. The most common group of bacteria in the oropharyngeal flora is viridans streptococci.[6] Bacterial species in the mouth are more diverse than those found in the upper or lower esophagus; it is notable that the prevalence of lactobacillus and enteric rods is highest in the distal esophagus, most likely because of the acidity of gastric contents.[6] In a large case series involving patients with an atrioesophageal fistula after ablation for atrial fibrillation, the most common organisms isolated were subspecies of viridans streptococci, followed by organisms such as *Mycoplasma salivarium*, enterococcus, and *Candida albicans*, findings that underscore the need for broad empirical treatment including coverage of gram-positive bacteria and fungal organisms in such cases.[7]

Blood cultures performed at the other hospital grew subspecies of viridans streptococci (*Streptococcus mitis* and *S. oralis*) and the anaerobe *Prevotella buccae* in all four bottles. Pleural-fluid cultures grew pseudomonas species. Pericardial-fluid cultures performed with samples obtained in the operating room at this hospital grew varied flora, predominantly subspecies of viridans streptococci (*S. anginosus*, *S. intermedius*, *S. constellatus*, *S. mitis*, and *S. oralis*), as well as lactobacillus, *Escherichia coli*, and *Rothia dentocariosa*. Pericardial-tissue cultures also grew multiple subspecies of viridans streptococci. After these results were received, the antimicrobial regimen was changed to a 4-week course of intravenous cefepime, metronidazole, fluconazole, and meropenem.

LABORATORY DIAGNOSIS

Esophageal–pericardial fistula.

FOLLOW-UP

Dr. Sachdeva: For the first 2 weeks after the procedure, the patient remained in critical condition in the surgical ICU. Trials of extubation were unsuccessful, and tracheostomy was performed, with subsequent discontinuation of ventilatory support. Drainage occurred through the nasogastric tube and the gastric port of the gastrojejunal tube, and total enteral nutrition was administered through the jejunal port. Chest and pericardial drains were removed sequentially.

Three weeks after the procedure, the patient was transferred out of the surgical ICU to the surgical floor, where she received physical therapy and speech and language therapy for rehabilitation. Decannulation was performed, and the nasogastric tube was removed. A barium swallow examination performed 8 weeks after the procedure showed no evidence of a leak at the surgical site. An oral diet was advanced slowly, and the gastrojejunal tube was capped.

The patient completed the 4-week course of cefepime, metronidazole, fluconazole, and meropenem for the treatment of sepsis and empyema. She continued to receive apixaban, amiodarone, and metoprolol for atrial fibrillation. The patient was discharged to a skilled nursing facility for 1 week and then returned home. At a follow-up visit 12 weeks after the procedure, she was able to maintain a diet of soft solid foods and to walk without assistance, and all tubes and catheters were removed. At a later follow-up visit, she had returned to her baseline functional status and active lifestyle.

FINAL DIAGNOSIS

Esophageal–pericardial fistula after percutaneous catheter ablation for atrial fibrillation.

This case was presented at the Wilkins Visiting Professorship lecture in the Thoracic Surgery Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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